

**Evaluating Expressions with Radicals**Question Bank - Additional 4.05 **ANSWER KEY**

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*Show your work. Circle your final answer.***For questions 1 through 6, evaluate.**

1.  $\sqrt{64}$

**8**

2.  $\sqrt[3]{8}$

**2**

3.  $\sqrt[3]{-125}$

**-5***An odd root keeps the sign of its radicand.*

4.  $\sqrt{\frac{9}{49}}$

 **$\frac{3}{7}$** *Top and bottom each get the root.*

5.  $\sqrt{4^2 + 33}$

**7** $16 + 33 = 49.$ 

6.  $\sqrt[3]{5^2 + 100}$

**5** $25 + 100 = 125.$

For questions 7 and 8, evaluate. Each expression has a power and a radical.

7. Evaluate each expression.

a.  $\sqrt{144} \cdot 3^{-2}$

$$\frac{4}{3}$$

$$12 \cdot \frac{1}{9}$$

b.  $\left(-\frac{1}{4}\right)^2 + \sqrt{3^2 + 27}$

$$\frac{97}{16}$$

$$\frac{1}{16} + 6$$

c.  $\sqrt[3]{-27} - 2^{-1}$

$$-\frac{7}{2}$$

$$-3 - \frac{1}{2}$$

8. Evaluate each expression.

a.  $\sqrt[3]{64} - 1.5(\sqrt{7^2 - 13})$

$$-5$$

$$4 - 1.5(6)$$

b.  $\left(-\frac{1}{2}\right)^3 - \sqrt[3]{4^2 + 11}$

$$-\frac{25}{8}$$

$$-\frac{1}{8} - 3$$

c.  $\sqrt{6^2 + 13} \cdot 2^{-3}$

$$\frac{7}{8}$$

$$7 \cdot \frac{1}{8}$$

### Working backwards.

9. Find the value of  $n$  that makes each statement true. (The radicand is the unknown here, so it is not one of the listed perfect values until you find it.)

a.  $\sqrt{n} = 15$

$n = 225$

b.  $\sqrt[3]{n} = -2$

$n = -8$

c.  $\sqrt{n-5} = 11$

$n = 126$

$n - 5 = 121$  .

### Reading someone else's work.

10. Deshawn writes  $\sqrt{100-36} = \sqrt{100} - \sqrt{36} = 4$  . His answer happens to be wrong by a small amount. Evaluate the expression correctly, and explain why his method cannot be trusted even when it lands close.

$\sqrt{100-36} = \sqrt{64} = 8$  . *The bar groups the subtraction, so it is finished before the root is taken; the root of a difference is not the difference of the roots, and how close his answer lands is an accident of the numbers.*

Ask him to try  $\sqrt{25-9}$  : his method gives 2 and the value is 4.

11. Which is the value of  $\sqrt{8^2 + 36}$  ?

☒ A. 10

☐ B. 100

☐ C. 50

☐ D. 14

A

12. Select ALL of the expressions whose value is 4.

☒ A.  $\sqrt{16}$

☒ B.  $\sqrt[3]{64}$

☒ C.  $\sqrt{5^2 - 9}$

☒ D.  $\sqrt[3]{3^3 + 37}$

☐ E.  $\sqrt[3]{-64}$

☐ F.  $\sqrt{2^3 + 17}$

A, B, C and D